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Acth-Producing Pituitary Carcinoma in the Falx Cerebri and Other Unusual Locations

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ACTH-producing Pituitary Carcinoma in the Falx Cerebri and other unusual locations Background: Pituitary carcinoma is a condition defined by metastasis of a pituitary tumor to a distant location. Pituitary carcinomas are very rare and account for approximately 0.12% of adenohypophyseal tumors (1). **Clinical case:** We present a case of a 29-year-old female who was followed up in our Endocrinology Department. Past medical history included the diagnosis of Cushing Disease and transsphenoidal tumor resection at 12 years of age, followed by transcranial resection two years later because of persistently elevated adrenocorticotropic hormone (ACTH). Despite the surgical management, the patient persisted with increased ACTH and hypercortisolism, so bilateral adrenalectomy was performed a year later. Two years after the procedure, the patient presented with a newly diagnosed pituitary macroadenoma, so the diagnosis of Nelson Syndrome was made. Linear accelerator radiotherapy was given, which reduced the size of the tumor. Later, multiple imaging studies showed multiple lesions on the falx cerebri, posterior clinoid process, retroclival region, cerebellopontine angle, pterygopalatine fossa, infratentorial region, and posterior ethmoidal cells. Biopsy and immunohistochemistry of the falx cerebri lesions described ACTH-producing pituitary adenocarcinoma. Treatment was given with intramuscular octreotide, dabrafenib, and trametinib. Despite persistently elevated ACTH levels, the patient has since remained clinically stable, without new development or worsening of symptoms. **Conclusion:** There are 3 unique aspects of our case. First, we reported an unusual presentation of this disease, because the patient in our case was a female with an early age of onset, in contrast with the reported literature. Second, this is the first reported case demonstrating pituitary carcinoma in the falx cerebri. Third, the prognosis of pituitary carcinoma is very poor, and mortality is extremely high, however, the patient in our case has been followed up for 7 years since the diagnosis of the metastatic lesions and has remained clinically stable. **References:** 1. Castellanos LE, Gutierrez C, Smith T, Laws ER, Iorgulescu JB. Epidemiology of common and uncommon adult pituitary tumors in the U.S. according to the 2017

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